LectureNotes Offline

Gujarati / Hindi lecture notes on Snapdragon, with no internet needed

Snapdragon® AI Lab Build & Present Challenge 2026

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The problem

Students struggle to take notes in mixed-language lectures.

Mixed languages

Lectures mix Gujarati, Hindi and English technical terms. Writing notes while listening is hard.

Needs internet

Most AI note tools need a steady connection. Campus Wi-Fi and mobile data are often unreliable.

Privacy and cost

Recordings are uploaded to cloud servers, using data and raising privacy concerns.

The solution

Load a recording. Get study material. Everything stays on your laptop.

1 Transcribe

Whisper turns speech into text in Gujarati, Hindi or English.

2 Summarize

A small on-device LLM (Llama 3.2 3B) writes short revision notes.

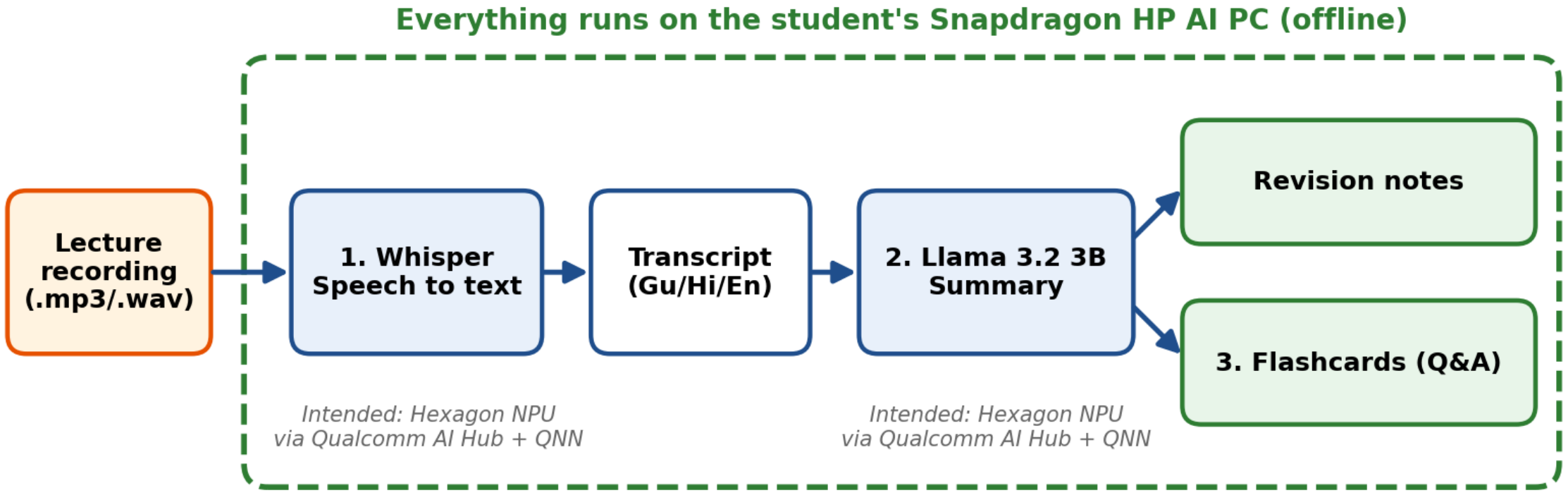
3 Flashcards

The same model creates question and answer cards.

4 Export

Transcript, Markdown notes and a flashcards CSV.

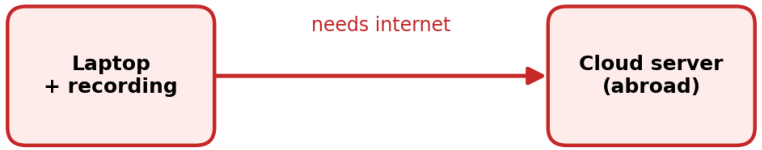
How it works



Whisper and Llama 3.2 3B are both available on Qualcomm AI Hub. The intended deployment compiles them for the Hexagon NPU and runs them through ONNX Runtime (QNN).

Why on-device on Snapdragon

Typical cloud note apps



Audio uploaded | data cost | privacy risk
often weaker regional-language support

LectureNotes Offline



Works offline | no data cost | private
Gujarati / Hindi / English

Works offline

No dependence on campus Wi-Fi or mobile data.

Private

Audio, transcripts and notes never leave the device.

Low power

The Hexagon NPU is designed to run these models efficiently, helping battery life.

What you get

Example output (illustrative, in English) - lecture: PN junction diode

NOTES

- A diode conducts easily in forward bias and blocks current in reverse bias.
- A rectifier uses diodes to convert AC into pulsating DC.
- Key terms: forward bias, reverse bias, rectifier

FLASHCARDS

Q: What does a diode do in reverse bias?

A: It blocks current.

Q: What is a rectifier used for?

A: Converting AC to DC.

Illustrative example only. Not measured output.

Status and honest limits

Current status

The prototype runs the full pipeline (transcribe, summarize, flashcards) on CPU. The NPU path is the intended deployment.

[EDIT: state whether you tested on a Snapdragon device and any measured results.]

Known limitations

- Gujarati transcription is less accurate than Hindi or English.
- A 3B model can write weaker Gujarati summaries.
- Whisper can mistake names and formulas.
- Review notes and cards before studying.

Next steps and thank you

Future work

- Compile and benchmark on a real Snapdragon HP AI PC
- Speaker separation for classroom questions
- Formula sheet extraction
- Export flashcards to Anki
- GTU subject glossaries (e.g. Basic Electronics)

Thank you

GitHub: **[paste your repository link]**

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